BANGALORE INSTITUTE OF TECHNOLOGY

An Autonomous Institution under Visvesvaraya Technological University, Belagavi
K. R. Road, V. V. Puram, Bengaluru - 560004

DEPARTMENT OF CSE/ISE/AI&ML/CSE(DS)/CSE(IOT)

MODULE-WISE QUESTION BANK (MODULE 1)

Academic Year :	2026-27	**Batch :**	2026
Course Code :	BCS501	**Course Title :**	Software Engineering & Project Management
Credits :	4	**Course Coordinator :**	Dr.KJ Bhanushree
Total Contact Hours :	4 hrs/week (L:T:P:S = 4:0:0:0)	**Total Hours of Pedagogy :**	50
Faculty In-charge :		**Semester & Section :**	5

Guidelines:

- Questions shall be mapped to the appropriate Course Outcome (CO).
- Bloom's cognitive level shall be indicated as L1 to L6, wherever applicable.
- Question category may be indicated as Conceptual / Numerical / Application / Analysis / Design / Case Study / Programming / Other, as applicable.
- The question bank shall provide adequate coverage of all modules and relevant COs.
- Minimum of 10 questions need to be included in each module.

Bloom's Taxonomy Reference

L1	L2	L3	L4	L5	L6
Remember	Understand	Apply	Analyze	Evaluate	Create

MODULE - 1: Software and Software Engineering, Process Models, Agile Development

Course Outcome: CO1 | Syllabus RBT ceiling: L2, L3

Q. No.	Question	CO	RBL	Category
1	Define software. List and explain the categories of software.	CO1	L2	Conceptual
2	Explain the unique nature of WebApps with their attributes.	CO1	L2	Conceptual
3	Describe the generic process model with its five framework activities and umbrella activities.	CO1	L2	Conceptual
4	Explain the Waterfall model with a neat diagram. Mention its advantages and drawbacks.	CO1	L2	Conceptual
5	Compare the Incremental and Evolutionary (Prototyping and Spiral) process models.	CO1	L2	Conceptual
6	Explain the Unified Process phases and their work products.	CO1	L2	Conceptual
7	Discuss the Personal Software Process (PSP) and Team Software Process (TSP).	CO1	L2	Conceptual
8	List and explain the software myths (management, customer, practitioner) and the realities behind them.	CO1	L2	Conceptual

9	What is agility? Explain how agility relates to the cost of change.	CO1	L2	Conceptual
10	With a neat diagram explain the Extreme Programming (XP) values, process, and practices.	CO1	L2	Conceptual
11	Briefly explain any three other agile process models: (Scrum, DSDM, Crystal, FDD, Lean Software Development)	CO1	L2	Conceptual
12	**Case Study 1 (Food-delivery app):** A food-delivery start-up must launch its mobile app in 8 weeks, and restaurant and customer feedback changes the requirements every week. Choose the most suitable process model (Waterfall, Incremental, or Agile/XP) and justify your choice against the alternatives.	CO1	L3	Case Study
13	**Case Study 2 (Hospital records system):** A hospital is building a patient-records system where regulations demand complete documentation and the requirements are fixed and well understood. The team proposes Agile with minimal documentation. Analyze whether this is appropriate and recommend a suitable prescriptive process model.	CO1	L3	Case Study
14	**Case Study 3 (Net-banking project):** A bank's net-banking project is in the testing phase when management asks for a new UPI payment feature, and the manager also plans to add more programmers to recover lost time. Using the cost-of-change curve, explain why this change is costly under Waterfall and how agility reduces that cost. Identify the software myth in the manager's plan.	CO1	L3	Case Study

Faculty In-charge

Course Coordinator Module Coordinator IQAC Program Coordinator